Gauri Makker

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Education

Vellore Institute of Technology (VIT) Bhopal Aug 2023 – May 2027
B.Tech in Computer Science and Engineering *CGPA: 8.44/10*
Coursework: Machine Learning, Foundations of Data Science, Design and Analysis of Algorithms, Applied Linear Algebra, Probability and Statistics, Parallel and Distributed Computing, Computer Networks

Publications

Makker, G. (2026). Where do slopes fail in Rudraprayag? A small-inventory landslide susceptibility study, and how evaluation choices change the answer. *EarthArXiv preprint.* doi.org/10.31223/X5KB9J

Experience

AI Engineer Intern, Neuronafence Feb 2025 – Jul 2025
- Built a retrieval-augmented generation (RAG) chatbot over 10,000+ internal company documents using Python, LangChain and Hugging Face Transformers.
- Designed the document ingestion pipeline (parsing, chunking, embedding) so the knowledge base could be updated without retraining.

Software Development and ML Intern, Caliche Global May 2024 – Jul 2024
- Built REST APIs in Flask for a farm management system, including an ML model that provides soil-health insights.
- Wrote SDLC documentation and worked in Agile sprints with the client team.

Projects

Landslide Susceptibility Mapping, Rudraprayag | *Python, Google Earth Engine, scikit-learn, Sentinel-2* GitHub
- Built an open landslide inventory from scratch: detected 330 candidate scars in Sentinel-2 and SRTM data and manually verified 300 on high-resolution imagery, confirming 38 landslides.
- Found the baseline random forest was mostly learning location (AUC 0.85 vs. 0.83 for a coordinates-only model); redesigned evaluation with spatial cross-validation, matched sampling and 20 random draws.
- Final 20-model ensemble places 87% of held-out landslides in its top two risk classes vs. 49% of nearby stable slopes; published as an interactive web map and preprint.

CommBot: Disaster Alerts by SMS | *Python, LLMs, pytest, CAP / NDMA SACHET* GitHub
- Converts official disaster warnings (NDMA, IMD) into Hindi or English SMS for basic phones without internet, within the 70-character Hindi SMS limit.
- Restricted the LLM to fixed labels and text copied from the source alert; any unverified fact is discarded, enforced by 35 automated tests.

PlayVision: Football Analytics with Computer Vision | *PyTorch, YOLOv8, ByteTrack, SigLIP, OpenCV*
- Team project: detected and tracked players, referees and the ball; identified teams without labels by clustering SigLIP embeddings and mapped positions to a top-down pitch view.

LawBot: RAG Legal Assistant | *LangChain, FAISS, Ollama, Streamlit* GitHub
- Team project: question-answering assistant restricted to uploaded legal PDFs, with prompts that refuse answers not grounded in the documents.

Game Engine and Path Tracer from Scratch | *C++17, CMake, OpenMP* GitHub
- 2D engine with ECS, A* pathfinding and procedural levels (29 test cases); physically based path tracer with BVH acceleration and multi-threading, no graphics libraries.

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL
ML and AI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, FAISS, Ollama, OpenCV, YOLOv8, RAG, NumPy, Pandas
Data and Backend: Flask, FastAPI, REST APIs, Streamlit, MySQL, Google Earth Engine
Tools: Git, GitHub, Linux, CMake, pytest, Jupyter Notebook, VS Code

Leadership and Awards

President, Notion Community Club, VIT Bhopal Jul 2025 – Mar 2026
- Led a 130+ member club; organized 10+ workshops and 2 hackathons with 300+ total participants.

Student Internship Coordinator, Institution's Innovation Council (IIC), VIT Bhopal Apr 2025 – Sep 2026
- Coordinated student participation in Ministry of Education innovation programs: startup ideation contests, IPR sessions and industry mentorship.

Certifications: NPTEL Introduction to Machine Learning (IIT Madras); NPTEL Marketing Analytics (IIT Kharagpur, Elite, 92%)
Awards: National finalist, INSPIRE-MANAK National Level Exhibition and Project Competition (2019), Government of India